

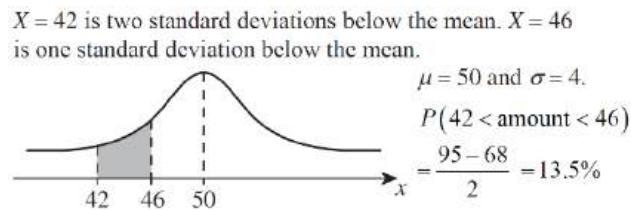


Continuous Probability Distribution Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 The amount of water that Eleanor uses to wash her car is normally distributed with a mean of 50 litres and a standard deviation of 4 litres. On what percentage of occasions would Eleanor expect to use between 42 litres and 46 litres of water to wash her car? 2



- 2 Let X be a normally distributed random variable with mean 6 and variance 9 and let Z be the random variable with the standard normal distribution.

- (i) Find $P(X > 6)$. $\sigma^2 = 9$ (or $\sigma = 3$) and $\mu = 6$ 1

$$P(X > 6) = P(X > \mu) = 0.5$$

- (ii) Find b such that $P(X > 8) = P(Z < b)$. 2

$$\begin{aligned} P(X > 8) &= P(X < 4) \text{ (symmetry)} \\ z &= \frac{x - \mu}{\sigma} = \frac{4 - 6}{3} = -\frac{2}{3} \\ P(X > 8) &= P\left(Z < -\frac{2}{3}\right) \therefore b = -\frac{2}{3} \end{aligned}$$

- 3 The probability density function for the continuous random variable X is given by:

$$f(x) = \begin{cases} kx(4-x) & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (i) What is the value of k ? 2

$$\int_1^4 kx(4-x) dx = 1$$

$$\int_1^4 4kx - kx^2 dx = 1$$

$$k \left[2x^2 - \frac{x^3}{3} \right]_1^4 = 1$$

$$k \left[\left(32 - \frac{64}{3} \right) - \left(2 - \frac{1}{3} \right) \right] = 1$$

$$9k = 1 \therefore k = \frac{1}{9}$$

- (ii) Find $F(t)$ the cumulative distribution function. 3

$$f(x) = \begin{cases} \frac{1}{9}x(4-x) & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

$$F(t) = \int_{-\infty}^t f(x) dx$$

$$= \int_1^t f(x) dx$$

$$= \int_1^t \frac{1}{9}x(4-x) dx = \frac{1}{9} \int_1^t 4x - x^2 dx$$

$$= \frac{1}{9} \left[2x^2 - \frac{x^3}{3} \right]_1^t$$

$$= \frac{1}{9} \left[\left(2t^2 - \frac{t^3}{3} \right) - \left(2 - \frac{1}{3} \right) \right] = \frac{1}{9} \left[2t^2 - \frac{t^3}{3} - \frac{5}{3} \right]$$

$$= \frac{1}{27} (6t^2 - t^3 - 5)$$

- 4 Pasta boxes are labelled with a net weight of 375 g. The business selling the pasta boxes fills the boxes to a net weight of 380 g, to avoid customer complaints. The weight of the pasta boxes follows a normal distribution with a mean weight of 380 g and a standard deviation of 2.5 g.

- (i) What is the probability that a box selected randomly will contain less than 375 g of pasta? 2

$$z = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{375 - 380}{2.5} = -2$$

95% of scores between -2 and 2.
Less than a z-score of -2.
 $5\% \div 2 = 2.5\%$
 $\therefore 2.5\%$ probability that a box has less than 375 g.

- (ii) The business sets a target of less than 16% of all boxes will contain more than 385 g of pasta. The business can change the mean weight of the boxes to meet this target. What is mean weight required to meet the target if the standard deviation remains at 2.5? 2

68% of scores between -1 and 1.
16% of scores are greater than 1. This is the target.

$$z = \frac{x - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$1 = \frac{385 - \mu}{2.5}$$

$$\mu = 385 - 2.5 = 382.5 \text{ g}$$

$\therefore$ Mean weight is 382.5 g.

- 5 The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & \text{for } x < 0, \\ \frac{1}{3}x^2(4 - x^2) & \text{for } 0 \leq x \leq 1, \\ 1 & \text{for } x > 1 \end{cases}$$

- (i) Find $P(X > 0.6)$. $P(X > 0.6) = 1 - F(0.6) = 1 - \frac{1}{3}(0.6)^2(4 - 0.6^2) = \frac{352}{625}$ or 0.5632 2

- (ii) Find the probability density function $f(x)$ of X . $f(x) = \frac{d}{dx}F(x) = \frac{d}{dx}\frac{1}{3}x^2(4 - x^2)$
 $= \frac{d}{dx}\frac{4}{3}x^2 - \frac{1}{3}x^4 = \frac{8x}{3} - \frac{4x^3}{3}$ for $0 \leq x \leq 1$. 1

- (iii) Determine the mode of the distribution. Give your answer correct to three decimal places. $f'(x) = \frac{4}{3}(2 - 3x^2) = 0$
Mode = $\sqrt{\frac{2}{3}} \approx 0.816496... = 0.816$ (3 decimal places) 1

- 6 Alison received her scores for a class test in each of two subjects. The class's scores for each test

Subject	Alison's score	Mean	Standard deviation
English	85	60	22
Mathematics	75	52	15

were normally distributed. The table shows the subjects and Alison's scores as well as the mean and standard deviation of the class scores on each test.

- (i) Calculate Alison's standardised z-scores for both tests. Give your answers correct to two decimal places. 2

English test: $z = \frac{x - \mu}{\sigma} = \frac{85 - 60}{22} \approx 1.14$ Mathematics test: $z = \frac{x - \mu}{\sigma} = \frac{75 - 52}{15} \approx 1.53$

- (ii) Alison is unsure if she performed better in the English test or the Mathematics test.

Relative to the rest of the class, in which subject has Alison achieved a better result? 2

Justify your answer with reference to the standardised z-scores calculated in part (a).

- 7 A continuous random variable T represents the number of days after contracting a virus. It has the following probability density function.

$$f(t) = \begin{cases} \frac{k}{2t-1} & \text{for } 1 \leq t \leq 14 \\ 0 & \text{for } 0 \leq t < 1 \text{ or } t > 14 \end{cases}$$

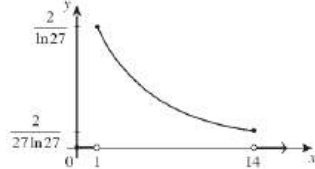
- (i) Show that $k = \frac{2}{\ln 27}$.

$$\int_1^{14} \frac{k}{2t-1} dt = 1 \quad \frac{k}{2} (\ln 27 - \ln 1) = 1$$

- (ii) On the axes below, sketch the graph of $y = f(t)$.

$$\frac{k}{2} \int_1^{14} \frac{2}{2t-1} dt = 1 \quad \frac{k}{2} (\ln 27) = 1$$

$$\frac{k}{2} [\ln|2t-1|]_1^{14} = 1 \quad k = \frac{2}{\ln 27}$$



- (iii) After how many days will a person have a 75% chance of showing symptoms after they have contracted the virus?

Let T be the time after symptoms of the virus first appear.

$$\int_1^T \frac{2}{\ln 27} \times \frac{1}{2t-1} dt = \frac{3}{4} \quad \frac{2}{\ln 27} \times \frac{1}{2} \int_1^T \frac{2}{2t-1} dt = \frac{3}{4}$$

$$\frac{1}{\ln 27} [\ln|2t-1|]_1^T = \frac{3}{4} \quad \ln|2T-1| = \frac{3 \ln 27}{4}$$

$$2T-1 = e^{\frac{3 \ln 27}{4}} \quad T = \frac{\left(e^{\frac{3 \ln 27}{4}}\right) + 1}{2} = 6.422 \dots \approx 7 \text{ days}$$

- 8 For a normal distribution, $P(X \leq 23.9) = 0.9192$ and $P(X \leq 17.15) = 0.3085$. Find the mean and standard deviation.

$$P(X \leq 23.9) = 0.9192$$

$$z = 1.4 \quad z = \frac{x - \mu}{\sigma}$$

$$1.4 = \frac{23.8 - \mu}{\sigma}$$

$$23.8 - \mu = 1.4\sigma$$

$$(2) - (1):$$

$$6.65 = 1.9\sigma$$

$$\sigma = 3.5$$

$$P(X \leq 17.15) = 0.3085$$

$$z = -0.5 \quad -0.5 = \frac{17.15 - \mu}{\sigma}$$

$$17.15 - \mu = -0.5\sigma \quad (1)$$

$$23.8 - \mu = 1.4\sigma \quad (2)$$

$$\text{Sub into (2):}$$

$$23.8 - \mu = 1.4 \times 3.5$$

$$\mu = 18.9$$



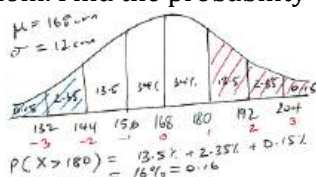
Continuous Probability Distribution Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 The heights of a group of students are normally distributed with a mean of 168 cm and a standard deviation of 12 cm.

- (i) A student is chosen at random. Find the probability that the student's height is greater than 180 cm. 1



- (ii) In this group of students, 2.5% have heights less than d . Find the value of d . 1

$$2.5\% = 0.025$$

$$0.025 = 0.0044 + 0.054$$

$$\therefore d = 144 \text{ cm}$$

- 2 A continuous random variable, X , has the following probability density function.

$$f(x) = \begin{cases} \sec^2 x & \text{for } 0 \leq x \leq p, \text{ where } p \text{ is a constant and } p < 1 \\ 0 & \text{for all other values of } x \end{cases}$$

$$\int_0^p \sec^2 x \, dx = 1$$

$$[\tan x]_0^p = 1$$

- (i) Calculate the value of p . $\tan p = 1$ $p = \frac{\pi}{4}$ 2
- (ii) Find $P(X \leq 0.2)$. Give your answer correct to five decimal places. 2

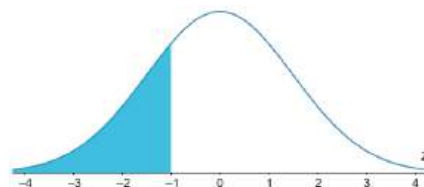
$$P(X \leq 0.2) = \int_0^{0.2} \sec^2 x \, dx$$

$$= [\tan x]_0^{0.2} = \tan 0.2 = 0.20271$$

- 3 The weight of chicken eggs is normally distributed with mean weight of 50 g and a standard deviation of 9 g. What percentage of eggs weigh between 41 g and 68 g? 1

Values range from 1 standard deviation below to 2 standard deviations above the mean.
From a normal distribution, we have approx. 34% of scores below the mean and 47.5% of scores above the mean. Total will be 81.5%

- 4 The graph shows the area under a normal distribution curve. Which of the following is equivalent to the shaded area? 1



A. $P(z \geq 1)$

B. $P(z \geq -1)$

C. $1 - P(-1 \leq z \leq 1)$

D. $P(z \leq 1)$

$$P(z \leq -1) = P(z \geq 1)$$

using symmetry

- 5 A cumulative distribution function is given by $F(x) = \frac{x^3 - 8}{335}$. Find the interquartile range of the continuous probability distribution. 2

$$Q_3 : 0.75 = \frac{x^3 - 8}{335}$$

$$251.25 = x^3 - 8$$

$$259.25 = x^3$$

$$x = 6.38 \text{ (2 dec. pl.)}$$

$$Q_1 : 0.25 = \frac{x^3 - 8}{335}$$

$$83.75 = x^3 - 8$$

$$91.75 = x^3$$

$$x = 4.51 \text{ (2 dec. pl.)}$$

$$IQR = Q_3 - Q_1 = 6.38 - 4.51 = 1.87$$

- 6 The length of time, that a customer queues in a Post Office is a random variable, T , with probability density function $f(t) = \begin{cases} c(81 - t^2), & 0 \leq t \leq 9 \\ 0, & \text{otherwise} \end{cases}$ where c is a constant.

(i) Show that the value of c is $\frac{1}{486}$. 2

$$\text{Since PDF } \int_0^9 c(81 - t^2) dt = 1$$

$$c \left[81t - \frac{t^3}{3} \right]_0^9 = 1$$

$$c(486) = 1$$

$$c = \frac{1}{486}$$

- (ii) Show that the cumulative distribution function $F(t)$ is given by 2

$$F(t) = \begin{cases} 0, & t < 0 \\ \frac{t}{6} - \frac{t^3}{1458}, & 0 \leq t \leq 9 \\ 1, & t > 9 \end{cases}$$

$$F(t) = \int_0^t \frac{1}{486} (81 - t^2) dt$$

$$= \frac{1}{486} \left[81t - \frac{t^3}{3} \right]_0^t$$

$$= \frac{1}{486} \left(81t - \frac{t^3}{3} \right)$$

- (iii) A customer has been queueing for 3 minutes. Find the probability that this customer will be queueing for at least 7 minutes. 3

$$\begin{aligned}
 P(T > 7 | T > 3) &= \frac{P(T > 7 \cap T > 3)}{P(T > 3)} = \frac{P(T > 7)}{P(T > 3)} \\
 &= \frac{1 - P(T \leq 7)}{1 - P(T \leq 3)} = \frac{1 - F(7)}{1 - F(3)} \\
 &= \frac{\frac{50}{729}}{\frac{14}{27}} = \frac{25}{189}
 \end{aligned}$$

- 7 (i) Show that the function $f(x) = \frac{\pi}{12} \sin\left(\frac{\pi x}{6}\right)$, $[0, 6]$ is a probability density function. 3

$$\begin{aligned}
 \int_0^6 \frac{\pi}{12} \sin\left(\frac{\pi x}{6}\right) dx &= \frac{\pi}{12} \left[-\frac{6}{\pi} \cos\left(\frac{\pi x}{6}\right) \right]_0^6 \\
 &= \frac{\pi}{12} \left[-\frac{6}{\pi} \cos \pi + \frac{6}{\pi} \cos 0 \right] = \frac{\pi}{12} \times \frac{12}{\pi} = 1
 \end{aligned}$$

$\therefore$ The function is a probability density function since $f(x) \geq 0$ and $\int_a^b f(x) dx = 1$.

- (ii) For a particular continuous random variable X , find $P(X \leq 4)$ for the function described in (i). 2

$$\begin{aligned}
 P(X \leq 4) &= \int_0^4 \frac{\pi}{12} \sin\left(\frac{\pi x}{6}\right) dx \\
 &= \frac{\pi}{12} \left[-\frac{6}{\pi} \cos\left(\frac{\pi x}{6}\right) \right]_0^4 \\
 &= \frac{\pi}{12} \times -\frac{6}{\pi} \times -\frac{1}{2} + \frac{\pi}{12} \times \frac{6}{\pi} = \frac{3}{4}
 \end{aligned}$$

- 8 A number of fish species are subject to minimum length regulations when they are caught. i.e. fish shorter than a given length must be returned to the water if caught. Two such species are Red Snapper and Barramundi which have minimum lengths of 30 cm and 55 cm respectively. A fishing tour operator in Northern Australia has made observations over a long period of time and found that, when measured in cm, both the variables 'R' (the lengths of caught Red Snapper) and 'B' (the lengths of caught Barramundi), are normally distributed. 'R' has a mean of 36 cm and standard deviation of 3 cm. 'B' has a standard deviation of 4 cm.

- (i) Calculate the mean length of Barramundi caught if 2.5% of Barramundi caught are less than 54 cm. Hence, $54 = \mu - 2s$ $\mu = 54 + 2 \times 4 = 62$ cm
ie. Mean length of caught Barramundi was 62 cm. 2

- (ii) Calculate the z-scores for the minimum allowed lengths for both variables, R and B, and comment upon what this means in terms of which of the two species are more likely to be returned to the water after capture. 3

$$R \text{ minimum length} = 30 \text{ cm} \quad B \text{ minimum length} = 55 \text{ cm}$$

$$z\text{-score} = \frac{x - \mu}{s} = \frac{30 - 36}{3} = -2 \quad z\text{-score} = \frac{x - \mu}{s} = \frac{55 - 62}{4} = -1.75$$

From the z-scores and empirical law, only 2.5% of caught Red Snapper will be returned to the water because they are too small (minimum length is 2 standard deviations from the mean), however, more than 2.5% of Barramundi will be returned to the water as minimum length is only 1.75 standard deviations from the mean. ie. caught Barramundi are more likely to be returned to the water

- 9 Historical data for a particular aptitude test show that its completion time has a mean of 5 minutes with a standard deviation of 30 seconds. As part of the selection process for an available job, an employer requires candidates to complete the test faster than 90% of all applicants to progress to the next stage. An extract from a probability table for the standard normal distribution is shown below. Darcy completed the aptitude test in 4 minutes and 23 seconds. Did Darcy qualify for the next stage of selection? Justify your answer by demonstrating your knowledge of the normal distribution and the application of z-scores to the problem. 3

z	second decimal place									
	+ .00	+ .01	+ .02	+ .03	+ .04	+ .05	+ .06	+ .07	+ .08	+ .09
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

From the table, the z-score for an applicant that is faster than 89.97% of the other applicants is 1.28. The z-score of an applicant that is faster than 90.15% of the other applicants is 1.29.

This mean qualifying time for next stage would be:

$$x = \mu - 1.28s \quad \text{OR} \quad x = \mu - 1.29s$$

$$= 5 \text{ minutes} - 1.28 \times 30 \text{ seconds} \quad = 5 \text{ minutes} - 1.29 \times 30 \text{ seconds}$$

$$= 4 \text{ minutes } 21.6 \text{ seconds} \quad = 4 \text{ minutes } 21.3 \text{ seconds}$$

(Either answer would be acceptable. 1.28 is closer to 90% probability, 1.29 ensures probability exceeds 90%)

Darcy's time of 4 minutes 23 seconds does not qualify her for the next stage of selection.

(This could also be justified by calculating Darcy's z-score. 37 s faster than the mean gives $z = 1.23$, less than the 1.28 required.)